Eigen-Séta speed-ups: results report

aburan28/eigenx#5 (merged 580b633). Measured on a shared 4-vCPU Xeon @ 2.10 GHz VM; security unchanged by construction and re-measured.

Decapsulation wall time (medians, measured)

set	main fc02c6b	PR #5	PR #5 + target-cpu=native
L1 (n=15)	1.231 s	1.070 s (1.15×)	0.953 s (1.29×)
L3 (n=6)	5.925 s	5.024 s (1.18×)	3.459 s (1.71×)
L5 (n=3)	13.01 s	11.58 s (1.12×)	7.567 s (1.72×)

Run run-20261017T2350Z-speedups (CARD-4). Ratios derived. Key generation is inconclusive (1–3 keys per set, dominated by the random attempt count); the per-attempt basis step measured 226 ⟶ 121 ms (2^{1078}) and 147 ⟶ 30 ms (3^{162}) at L1.

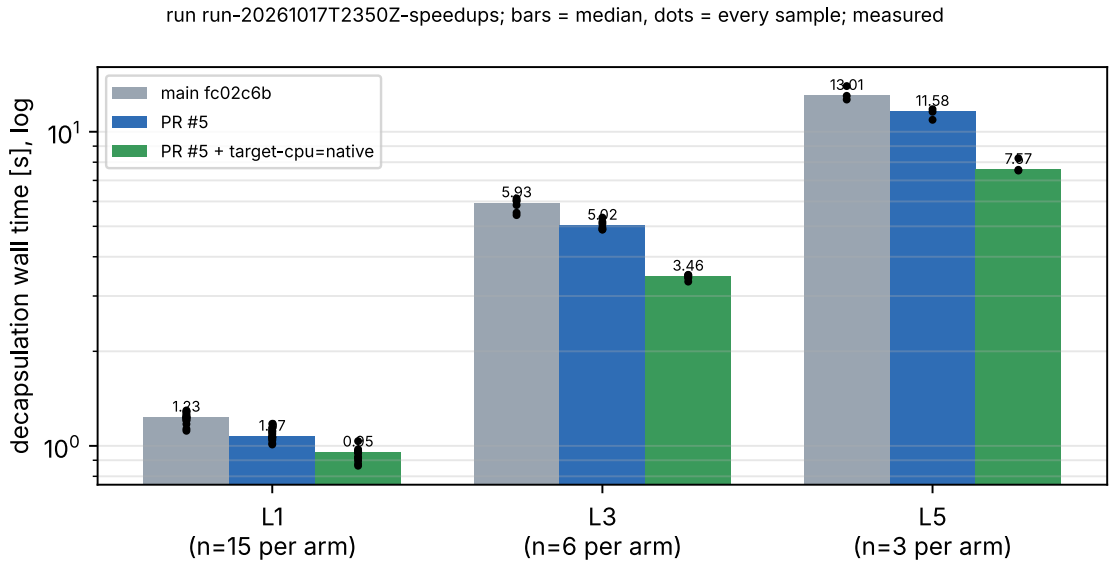


Figure 1: Bars median, dots every sample, log scale; measured.

Constant time and equivalence (measured)

Callgrind, all threads, identical inputs written by the main build: secret-phase instructions $18.18 \cdot 10^9$ ⟶ $12.36 \cdot 10^9$ (−32.0%); spread over two keys × valid/FO-rejected 286,937 ⟶ 2,027 Ir. Shared secrets byte-identical across builds; security_estimate output byte-identical.

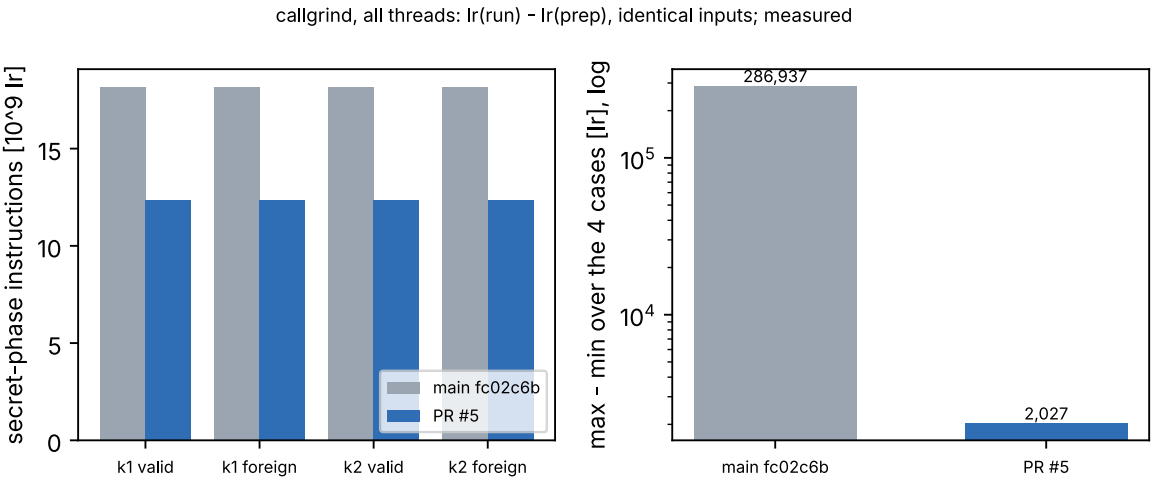


Figure 2: Ir(run) − Ir(prep); measured.

Not taken

Hand-written asm field multiply (needs `unsafe`; no faster than a native build). Half-length chains on a smaller prime: feasibility unproven, speed-up 1×–2.6×, torsion margin over $\log_2$ (d·f) would fall from about 540 to about 6 bits (ITEM7_HALF_CHAINS.md).

Distance to practicality

L1 decapsulation is 0.95–1.07 s; a practical PKE needs tens of milliseconds. Next: the theta chain step (618–670 F_{p^2} multiplications measured vs about 256 modelled), native builds, an AVX-512 IFMA multiplier in safe Rust, Karatsuba and lazy reduction.